

ABCPMX-II

1.00 beta 2

CALCULATION REPORT

Project Pr#1

Beam B#1



User :
Company :
Project : Project Pr#1

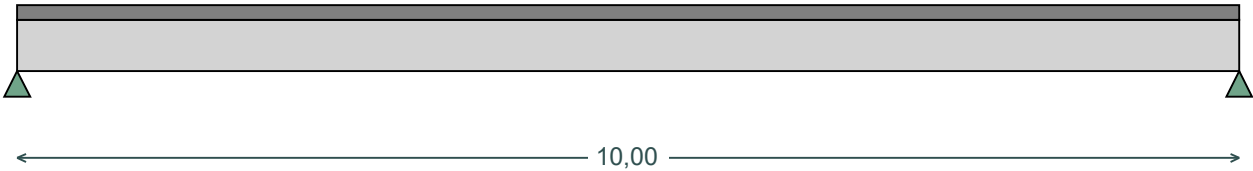
I - Parameters

Warning: This calculation is carried out in the Expert mode.
Some parameters may be outside the scope of the software

I.1 - Main parameters

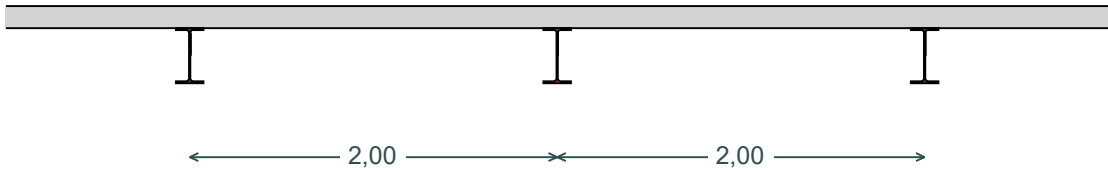
- Longitudinal parameters

Type of beam :	Composite beam
Total length of the beam :	$L_{tot} = 10,00$ m
Number of spans :	1
Type of beam :	Composite
Type of slab :	Composite



- Transversal parameters

Position of the beam :	Intermediate beam
Left spacing :	$d_1 = 2,00$ m
Right spacing :	$d_2 = 2,00$ m
No slab opening	



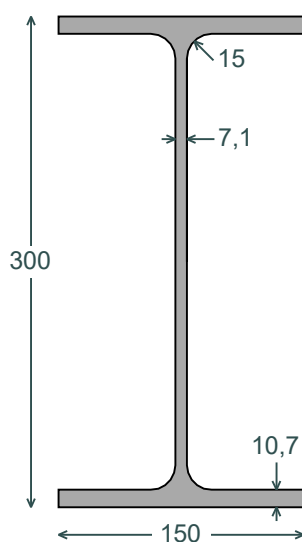
I.2 - Steel profile

- General

Type of steel profile : Hot rolled profile
 Name of the steel profile : IPE 300

- Dimensions

Total height of the section : $h_s = 300$ mm
 Width of the flanges : $b_f = 150$ mm
 Thickness of the flanges : $t_f = 10,7$ mm
 Height of the web (between flanges) : $h_w = 279$ mm
 Height of the web (between fillets) : $d_w = 249$ mm
 Thickness of the web : $t_w = 7,1$ mm
 Radius of the root fillet : $r = 15$ mm



- Properties

Cross-sectional area : $A = 53,81$ cm²
 Shear area : $A_v = 25,68$ cm²
 Second moment or area (y-y axis) : $I_y = 8356$ cm⁴
 Location of the centroid : $Z_G = 150$ mm
 Radius of gyration : $i_y = 124,6$ mm
 Second moment or area (z-z axis) : $I_z = 603$ cm⁴
 Elastic section modulus (y-y axis) : $W_{el,y} = 557,1$ cm³
 Elastic section modulus (z-z axis) : $W_{el,z} = 80,4$ cm³
 Plastic section modulus (y-y axis) : $W_{pl,y} = 628,4$ cm³
 Plastic section modulus (z-z axis) : $W_{pl,z} = 123,9$ cm³
 Torsional constant : $I_t = 19,92$ cm⁴
 Warping constant : $I_w = 125934$ cm⁶

Mass by length unit : $m_{lin} = 42,2$ kg/m
 Mass : $m = 422,4$ kg
 Total paint surface : $S = 11,60$ m²
 Paint surface (without upper face) : $S = 10,10$ m²
 Section factor : $M = 215,6$ m⁻¹
 Section factor (without upper face) : $M' = 187,7$ m⁻¹

- Material: steel

Steel grade :	S355 M/ML
Standard :	EN 10025-4
Nominal yield strength :	$f_y = 355 \text{ MPa}$
Ultimate strength :	$f_u = 470 \text{ MPa}$
Young's modulus :	$E = 210000 \text{ MPa}$
Density :	$\rho = 7850 \text{ kg/m}^3$

- Profile resistances

Axial resistance :	$N_{t,Rd} = 1910 \text{ kN}$
Shear resistance :	$V_{pl,Rd} = 526 \text{ kN}$
Plastic resistance to bending :	$M_{pl,Rd} = 223 \text{ kN.m}$
Location of the plastic neutral axis :	$z_{ANP} = 150 \text{ mm}$
Elastic resistance to bending :	$M_{el,Rd} = 198 \text{ kN.m}$
Section class for pure bending :	1
Section class for pure compression :	4
Shear buckling resistance :	$h_w/t_w = 39,2 \leq 72\varepsilon/\eta = 58,6$ Verification of web buckling not required

I.3 - Slab**- Geometry of the slab**

Type :	Composite slab
Thickness of the slab :	$t_d = 120 \text{ mm}$
Thickness above the deck :	$t_c = 62 \text{ mm}$
Notional size (EN 1992-1-1 Annex B) :	$h_0 = 124 \text{ mm}$
The slab is not continuous at the left support	
The slab is not continuous at the right support	

- Material: Concrete

Type :	Normal weight concrete
Class of concrete :	C20/25
Characteristic value of the compression strength :	$f_{ck} = 20 \text{ MPa}$
Mean value of compression strength :	$f_{cm} = 28,0 \text{ MPa}$
Mean tensile strength :	$f_{ctm} = 2,21 \text{ MPa}$
Mean secant modulus of elasticity :	$E_{cm} = 29962 \text{ MPa}$

- Longitudinal reinforcements

Number of layers :	1
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i	$e_{si} \text{ (mm)}$	$d_{si} \text{ (mm)}$	$z_{si} \text{ (mm)}$	$A_{si} \text{ (mm}^2\text{/m)}$
1	200	12	32	565

With

 e_{si} : Spacing of rebars axes d_{si} : Diameter of the rebars z_{si} : Location of the layer axis, measured from the upper face of the slab A_{si} : Area of steel rebars per unit length

- Material: Steel reinforcements

Class :

Yield strength :

Young's modulus :

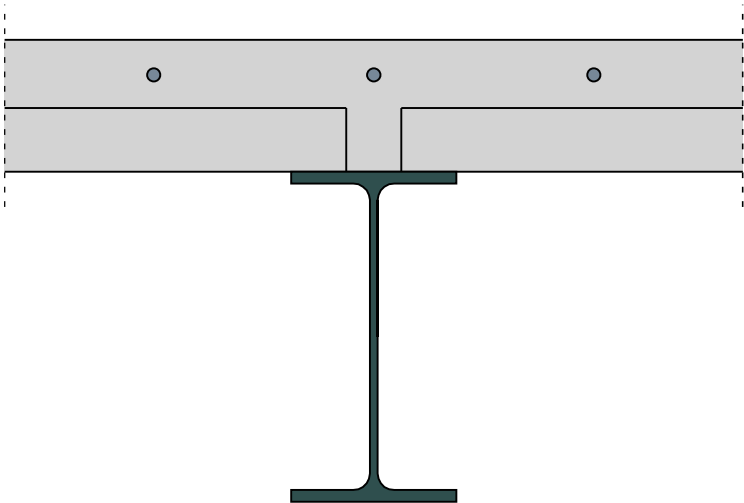
B500

f_{sk}

= 500 MPa

E_s

= 210000 MPa



- Profiled steel sheeting

Orientation of the profiled steel sheeting :

Profiled steel sheeting from :

Name :

Name :

Transversal

Database

Cofraplus_60 1.00

ArcelorMittal / Cofraplus_60 1.00

Dimensions and properties:

Thickness :

Spacing between ribs axes :

Height of the ribs :

Bottom width of ribs :

Top width of ribs :

Surface mass :

Yield strength :

Second moment of area per unit of width :

Configuration of the ribs on support :

t_p

= 1,00 mm

e_p

= 207 mm

h_p

= 58 mm

b_b

= 62 mm

b_t

= 101 mm

μ_p

= 11,37 kg/m²

f_p

= 350 MPa

I_{pu}

= 60,08 cm⁴/m

Concrete rib and steel sheeting interrupted

- Connectors

Height of the stud :	h_{sc}	= 100 mm
Diameter of the shank :	d	= 19 mm
Yield strength :	f_{ysc}	= 350 MPa
Ultimate strength :	f_{usc}	= 450 MPa
Shear resistance :	P_{Rd}	= 46,2 kN ($n_r = 1$)
Reduction factor :	k_t	= 0,71 ($n_r = 1$)

- Connection arrangement

Span	Zone length (m)	Number of rows	Disposition every	Longi. spacing (mm)
1	10	1	1 Rib	200

I.4 - Lateral restraints

Type of restraint : No intermediate restraint

I.5 - Propping

Type of propping : Unpropped

I.6 - Loads

List of load cases defined by the User

- G1 : Self-weight**Distributed load**

x (m)	F (kN/m)	x (m)	F (kN/m)
0	4,80	10,00	4,80

I.7 - Partial factors**- Loads**

$\gamma_{G,sup}$	= 1,35
$\gamma_{G,inf}$	= 1,00
γ_Q	= 1,50

- Combination factors

$\psi_{0,Q1}$	= 0,7	$\psi_{0,Q2}$	= 0,7
$\psi_{1,Q1}$	= 0,5	$\psi_{1,Q2}$	= 0,5
$\psi_{2,Q1}$	= 0,3	$\psi_{2,Q2}$	= 0,3

- Resistance

Steel resistance		Slab resistance		Fire resistance	
γ_{M0}	= 1,00	γ_C	= 1,50	$\gamma_{M,fi}$	= 1,00
γ_{M1}	= 1,00	γ_s	= 1,15	$\gamma_{C,fi}$	= 1,00
γ_{M2}	= 1,25	γ_V	= 1,25	$\gamma_{s,fi}$	= 1,00
		γ_P	= 1,00	$\gamma_{V,fi}$	= 1,00

1.8 - Combinations

- Ultimate limit states

ULS-01 = 1,35 G1

- Serviceability limit states

SLS-01 = 1,00 G1 + 1,00 SHC

- Construction stage ultimate limit states

ULS-C-01 = 1,35 G1

- Construction stage serviceability limit states

SLS-C-01 = 1,00 G1

1.9 - Calculation parameters

- Standard

Standard :

First generation of Eurocodes

- Options at Ultimate Limit States

Calculation type :

Design of sections according to their class

Shear buckling :

$\eta = 1,00$

- Options at Serviceability Limit States

The deflections of composite beams take into account connectors stiffness

Slip of the connector at 0,7 PRk (prEN 1994-1-1 B.2.5 (5)) :

$s_e = 0,50 \text{ mm}$

Control of concrete cracking :

No

Limit of crack width. :

$w_{\max} = 0,40 \text{ mm}$

- Properties of concrete

Relative humidity :

RH = 50,00 %

Deformation of concrete shrinkage :

$\epsilon_{sh} = 325,0 \times 10^{-6}$

Young's modulus of rebars :

$E_s = 210000 \text{ MPa}$

- Properties of cross-sections

Simplified distribution of slab effective width :

No

Rebars in compression taken into account :

No

Age of concrete at loading	Slab	Encasement
Self-weight	28 d	56 d
Other permanent loads	28 d	56 d
Shrinkage	1 d	1 d

Age of concrete :

$t = 18250 \text{ d}$

- Parameters

G-force :

$g = 9,81 \text{ m/s}^2$

II - Sections properties

Notes:

z locations are about the interface between the profile and the slab

II.1 - Properties at mid span ($M > 0$)

Effective width of the slab : $b_{\text{eff}} = 2,00 \text{ m}$

- Elastic properties

n slab	$I_{\text{el},y}$ cm^4	$z_{\text{el},y}$ mm
-	8356	-150
7,01	32492	33,3
29,1	22067	-44,5
30,8	21644	-47,7

- Plastic properties

Plastic moment : $M_{\text{pl,Rd}} = 410 \text{ kN.m}$
 Location of the plastic neutral axis : $z_{\text{pl}} = -4,7 \text{ mm}$
 Plastic moment (characteristic value) : $M_{\text{pl,Rk}} = 462 \text{ kN.m}$

III - Analysis - Internal forces and moments

III.1 - Load cases in the analysis of the beam

Load cases	Stage	n Slab
G1 Self weight	Steel	-
SHC Shrinkage	Composite	29,1

III.2 - Load cases

- G1 : Self-weight

Reactions at supports :

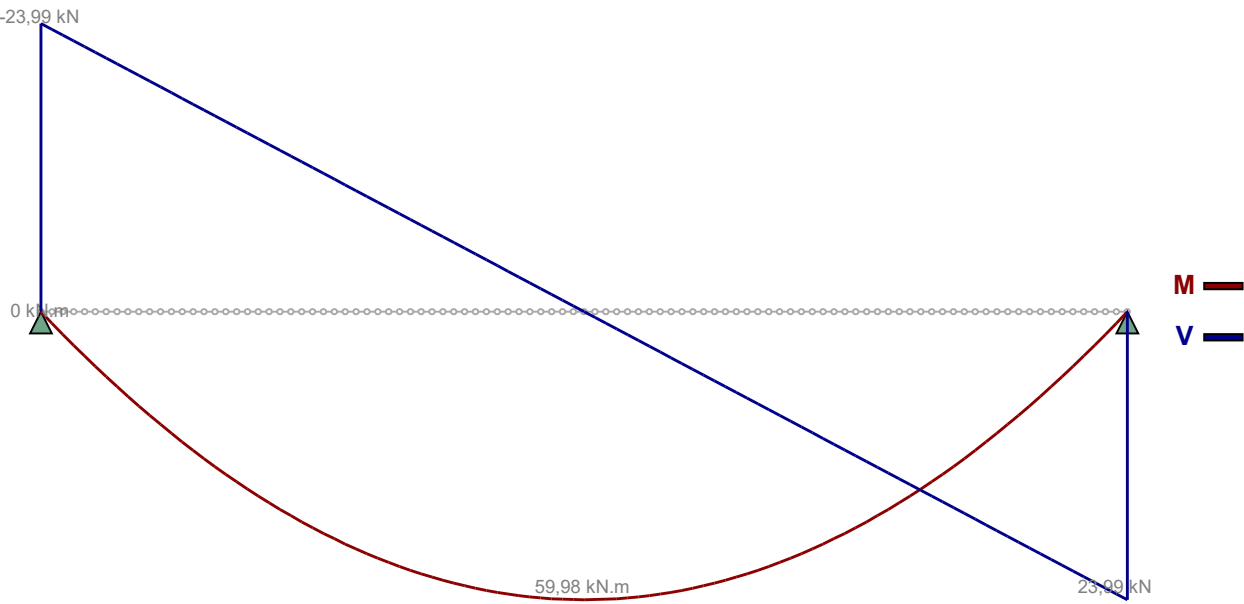
$$R_A = 24,0 \text{ kN}$$

$$R_B = 24,0 \text{ kN}$$

Node	x (m)	V_g (kN)	V_d (kN)	M_g (kN.m)	M_d (kN.m)
1	0		-23,99		0
2	0,10	-23,51		2,375	
3	0,20	-23,03		4,703	
4	0,30	-22,55		6,982	
5	0,40	-22,07		9,213	
6	0,50	-21,59		11,40	
7	0,60	-21,11		13,53	
8	0,70	-20,63		15,62	
9	0,80	-20,15		17,66	
10	0,90	-19,67		19,65	
11	1,00	-19,19		21,59	
12	1,10	-18,71		23,49	
13	1,20	-18,23		25,34	
14	1,30	-17,76		27,14	
15	1,40	-17,28		28,89	
16	1,50	-16,80		30,59	
17	1,60	-16,32		32,25	
18	1,70	-15,84		33,85	
19	1,80	-15,36		35,41	
20	1,90	-14,88		36,93	
21	2,00	-14,40		38,39	
22	2,10	-13,92		39,80	
23	2,20	-13,44		41,17	
24	2,30	-12,96		42,49	
25	2,40	-12,48		43,76	
26	2,50	-12,00		44,99	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
27	2,60	-11,52		46,16	
28	2,70	-11,04		47,29	
29	2,80	-10,56		48,37	
30	2,90	-10,08		49,40	
31	3,00	-9,597		50,39	
32	3,10	-9,117		51,32	
33	3,20	-8,638		52,21	
34	3,30	-8,158		53,05	
35	3,40	-7,678		53,84	
36	3,50	-7,198		54,58	
37	3,60	-6,718		55,28	
38	3,70	-6,238		55,93	
39	3,80	-5,758		56,53	
40	3,90	-5,279		57,08	
41	4,00	-4,799		57,58	
42	4,10	-4,319		58,04	
43	4,20	-3,839		58,45	
44	4,30	-3,359		58,81	
45	4,40	-2,879		59,12	
46	4,50	-2,399		59,38	
47	4,60	-1,919		59,60	
48	4,70	-1,440		59,77	
49	4,80	-0,960		59,89	
50	4,90	-0,480		59,96	
51	5,00	0,000		59,98	
52	5,10	0,480		59,96	
53	5,20	0,960		59,89	
54	5,30	1,440		59,77	
55	5,40	1,919		59,60	
56	5,50	2,399		59,38	
57	5,60	2,879		59,12	
58	5,70	3,359		58,81	
59	5,80	3,839		58,45	
60	5,90	4,319		58,04	
61	6,00	4,799		57,58	
62	6,10	5,279		57,08	
63	6,20	5,758		56,53	
64	6,30	6,238		55,93	
65	6,40	6,718		55,28	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
66	6,50	7,198		54,58	
67	6,60	7,678		53,84	
68	6,70	8,158		53,05	
69	6,80	8,638		52,21	
70	6,90	9,117		51,32	
71	7,00	9,597		50,39	
72	7,10	10,08		49,40	
73	7,20	10,56		48,37	
74	7,30	11,04		47,29	
75	7,40	11,52		46,16	
76	7,50	12,00		44,99	
77	7,60	12,48		43,76	
78	7,70	12,96		42,49	
79	7,80	13,44		41,17	
80	7,90	13,92		39,80	
81	8,00	14,40		38,39	
82	8,10	14,88		36,93	
83	8,20	15,36		35,41	
84	8,30	15,84		33,85	
85	8,40	16,32		32,25	
86	8,50	16,80		30,59	
87	8,60	17,28		28,89	
88	8,70	17,76		27,14	
89	8,80	18,23		25,34	
90	8,90	18,71		23,49	
91	9,00	19,19		21,59	
92	9,10	19,67		19,65	
93	9,20	20,15		17,66	
94	9,30	20,63		15,62	
95	9,40	21,11		13,53	
96	9,50	21,59		11,40	
97	9,60	22,07		9,213	
98	9,70	22,55		6,982	
99	9,80	23,03		4,703	
100	9,90	23,51		2,375	
101	10,00	23,99		0	



- SHC : Shrinkage of the slab

Reactions at supports :

$R_A = 0,0 \text{ kN}$
 $R_B = 0,0 \text{ kN}$

Node	x (m)	V_g (kN)	V_d (kN)	M_g (kN.m)	M_d (kN.m)
1	0		0,000		38,77
2	0,10	0,000		38,77	
3	0,20	0,000		38,77	
4	0,30	0,000		38,77	
5	0,40	0,000		38,77	
6	0,50	0,000		38,77	
7	0,60	0,000		38,77	
8	0,70	0,000		38,77	
9	0,80	0,000		38,77	
10	0,90	0,000		38,77	
11	1,00	0,000		38,77	
12	1,10	0,000		38,77	
13	1,20	0,000		38,77	
14	1,30	0,000		38,77	
15	1,40	0,000		38,77	
16	1,50	0,000		38,77	
17	1,60	0,000		38,77	
18	1,70	0,000		38,77	
19	1,80	0,000		38,77	
20	1,90	0,000		38,77	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
21	2,00	0,000		38,77	
22	2,10	0,000		38,77	
23	2,20	0,000		38,77	
24	2,30	0,000		38,77	
25	2,40	0,000		38,77	
26	2,50	0,000		38,77	
27	2,60	0,000		38,77	
28	2,70	0,000		38,77	
29	2,80	0,000		38,77	
30	2,90	0,000		38,77	
31	3,00	0,000		38,77	
32	3,10	0,000		38,77	
33	3,20	0,000		38,77	
34	3,30	0,000		38,77	
35	3,40	0,000		38,77	
36	3,50	0,000		38,77	
37	3,60	0,000		38,77	
38	3,70	0,000		38,77	
39	3,80	0,000		38,77	
40	3,90	0,000		38,77	
41	4,00	0,000		38,77	
42	4,10	0,000		38,77	
43	4,20	0,000		38,77	
44	4,30	0,000		38,77	
45	4,40	0,000		38,77	
46	4,50	0,000		38,77	
47	4,60	0,000		38,77	
48	4,70	0,000		38,77	
49	4,80	0,000		38,77	
50	4,90	0,000		38,77	
51	5,00	0,000		38,77	
52	5,10	0,000		38,77	
53	5,20	0,000		38,77	
54	5,30	0,000		38,77	
55	5,40	0,000		38,77	
56	5,50	0,000		38,77	
57	5,60	0,000		38,77	
58	5,70	0,000		38,77	
59	5,80	0,000		38,77	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
60	5,90	0,000		38,77	
61	6,00	0,000		38,77	
62	6,10	0,000		38,77	
63	6,20	0,000		38,77	
64	6,30	0,000		38,77	
65	6,40	0,000		38,77	
66	6,50	0,000		38,77	
67	6,60	0,000		38,77	
68	6,70	0,000		38,77	
69	6,80	0,000		38,77	
70	6,90	0,000		38,77	
71	7,00	0,000		38,77	
72	7,10	0,000		38,77	
73	7,20	0,000		38,77	
74	7,30	0,000		38,77	
75	7,40	0,000		38,77	
76	7,50	0,000		38,77	
77	7,60	0,000		38,77	
78	7,70	0,000		38,77	
79	7,80	0,000		38,77	
80	7,90	0,000		38,77	
81	8,00	0,000		38,77	
82	8,10	0,000		38,77	
83	8,20	0,000		38,77	
84	8,30	0,000		38,77	
85	8,40	0,000		38,77	
86	8,50	0,000		38,77	
87	8,60	0,000		38,77	
88	8,70	0,000		38,77	
89	8,80	0,000		38,77	
90	8,90	0,000		38,77	
91	9,00	0,000		38,77	
92	9,10	0,000		38,77	
93	9,20	0,000		38,77	
94	9,30	0,000		38,77	
95	9,40	0,000		38,77	
96	9,50	0,000		38,77	
97	9,60	0,000		38,77	
98	9,70	0,000		38,77	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
99	9,80	0,000		38,77	
100	9,90	0,000		38,77	
101	10,00	0,000		38,77	



III.3 - ULS Combinations

- ULS-01

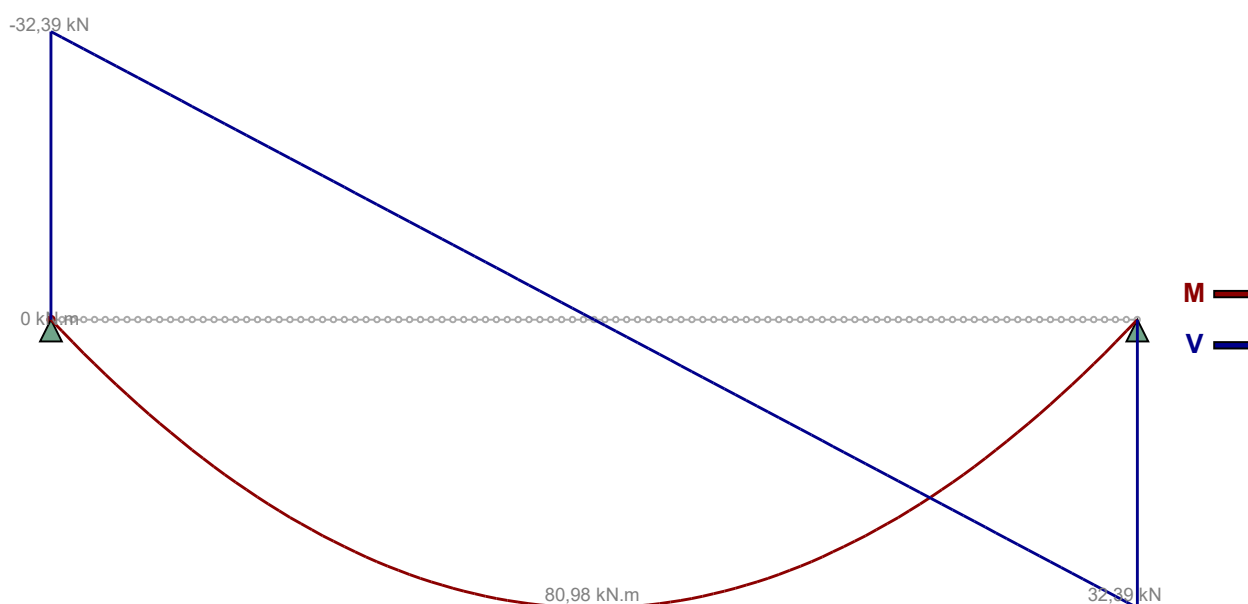
ULS-01 = 1,35 G1

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
1	0		-32,39		0
2	0,10	-31,74		3,207	
3	0,20	-31,10		6,349	
4	0,30	-30,45		9,426	
5	0,40	-29,80		12,44	
6	0,50	-29,15		15,39	
7	0,60	-28,50		18,27	
8	0,70	-27,86		21,09	
9	0,80	-27,21		23,84	
10	0,90	-26,56		26,53	
11	1,00	-25,91		29,15	
12	1,10	-25,27		31,71	
13	1,20	-24,62		34,20	
14	1,30	-23,97		36,63	
15	1,40	-23,32		39,00	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
16	1,50	-22,67		41,30	
17	1,60	-22,03		43,53	
18	1,70	-21,38		45,70	
19	1,80	-20,73		47,81	
20	1,90	-20,08		49,85	
21	2,00	-19,43		51,83	
22	2,10	-18,79		53,74	
23	2,20	-18,14		55,58	
24	2,30	-17,49		57,36	
25	2,40	-16,84		59,08	
26	2,50	-16,20		60,73	
27	2,60	-15,55		62,32	
28	2,70	-14,90		63,84	
29	2,80	-14,25		65,30	
30	2,90	-13,60		66,69	
31	3,00	-12,96		68,02	
32	3,10	-12,31		69,28	
33	3,20	-11,66		70,48	
34	3,30	-11,01		71,62	
35	3,40	-10,37		72,69	
36	3,50	-9,717		73,69	
37	3,60	-9,069		74,63	
38	3,70	-8,422		75,50	
39	3,80	-7,774		76,31	
40	3,90	-7,126		77,06	
41	4,00	-6,478		77,74	
42	4,10	-5,830		78,35	
43	4,20	-5,183		78,90	
44	4,30	-4,535		79,39	
45	4,40	-3,887		79,81	
46	4,50	-3,239		80,17	
47	4,60	-2,591		80,46	
48	4,70	-1,943		80,69	
49	4,80	-1,296		80,85	
50	4,90	-0,648		80,95	
51	5,00	0,000		80,98	
52	5,10	0,648		80,95	
53	5,20	1,296		80,85	
54	5,30	1,943		80,69	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
55	5,40	2,591		80,46	
56	5,50	3,239		80,17	
57	5,60	3,887		79,81	
58	5,70	4,535		79,39	
59	5,80	5,183		78,90	
60	5,90	5,830		78,35	
61	6,00	6,478		77,74	
62	6,10	7,126		77,06	
63	6,20	7,774		76,31	
64	6,30	8,422		75,50	
65	6,40	9,069		74,63	
66	6,50	9,717		73,69	
67	6,60	10,37		72,69	
68	6,70	11,01		71,62	
69	6,80	11,66		70,48	
70	6,90	12,31		69,28	
71	7,00	12,96		68,02	
72	7,10	13,60		66,69	
73	7,20	14,25		65,30	
74	7,30	14,90		63,84	
75	7,40	15,55		62,32	
76	7,50	16,20		60,73	
77	7,60	16,84		59,08	
78	7,70	17,49		57,36	
79	7,80	18,14		55,58	
80	7,90	18,79		53,74	
81	8,00	19,43		51,83	
82	8,10	20,08		49,85	
83	8,20	20,73		47,81	
84	8,30	21,38		45,70	
85	8,40	22,03		43,53	
86	8,50	22,67		41,30	
87	8,60	23,32		39,00	
88	8,70	23,97		36,63	
89	8,80	24,62		34,20	
90	8,90	25,27		31,71	
91	9,00	25,91		29,15	
92	9,10	26,56		26,53	
93	9,20	27,21		23,84	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
94	9,30	27,86		21,09	
95	9,40	28,50		18,27	
96	9,50	29,15		15,39	
97	9,60	29,80		12,44	
98	9,70	30,45		9,426	
99	9,80	31,10		6,349	
100	9,90	31,74		3,207	
101	10,00	32,39		0	



III.4 - ULS Combinations at the construction stage

- ULS-C-01

ULS-C-01 = 1,35 G1

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
1	0		-32,39		0
2	0,10	-31,74		3,207	
3	0,20	-31,10		6,349	
4	0,30	-30,45		9,426	
5	0,40	-29,80		12,44	
6	0,50	-29,15		15,39	
7	0,60	-28,50		18,27	
8	0,70	-27,86		21,09	
9	0,80	-27,21		23,84	
10	0,90	-26,56		26,53	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
11	1,00	-25,91		29,15	
12	1,10	-25,27		31,71	
13	1,20	-24,62		34,20	
14	1,30	-23,97		36,63	
15	1,40	-23,32		39,00	
16	1,50	-22,67		41,30	
17	1,60	-22,03		43,53	
18	1,70	-21,38		45,70	
19	1,80	-20,73		47,81	
20	1,90	-20,08		49,85	
21	2,00	-19,43		51,83	
22	2,10	-18,79		53,74	
23	2,20	-18,14		55,58	
24	2,30	-17,49		57,36	
25	2,40	-16,84		59,08	
26	2,50	-16,20		60,73	
27	2,60	-15,55		62,32	
28	2,70	-14,90		63,84	
29	2,80	-14,25		65,30	
30	2,90	-13,60		66,69	
31	3,00	-12,96		68,02	
32	3,10	-12,31		69,28	
33	3,20	-11,66		70,48	
34	3,30	-11,01		71,62	
35	3,40	-10,37		72,69	
36	3,50	-9,717		73,69	
37	3,60	-9,069		74,63	
38	3,70	-8,422		75,50	
39	3,80	-7,774		76,31	
40	3,90	-7,126		77,06	
41	4,00	-6,478		77,74	
42	4,10	-5,830		78,35	
43	4,20	-5,183		78,90	
44	4,30	-4,535		79,39	
45	4,40	-3,887		79,81	
46	4,50	-3,239		80,17	
47	4,60	-2,591		80,46	
48	4,70	-1,943		80,69	
49	4,80	-1,296		80,85	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
50	4,90	-0,648		80,95	
51	5,00	0,000		80,98	
52	5,10	0,648		80,95	
53	5,20	1,296		80,85	
54	5,30	1,943		80,69	
55	5,40	2,591		80,46	
56	5,50	3,239		80,17	
57	5,60	3,887		79,81	
58	5,70	4,535		79,39	
59	5,80	5,183		78,90	
60	5,90	5,830		78,35	
61	6,00	6,478		77,74	
62	6,10	7,126		77,06	
63	6,20	7,774		76,31	
64	6,30	8,422		75,50	
65	6,40	9,069		74,63	
66	6,50	9,717		73,69	
67	6,60	10,37		72,69	
68	6,70	11,01		71,62	
69	6,80	11,66		70,48	
70	6,90	12,31		69,28	
71	7,00	12,96		68,02	
72	7,10	13,60		66,69	
73	7,20	14,25		65,30	
74	7,30	14,90		63,84	
75	7,40	15,55		62,32	
76	7,50	16,20		60,73	
77	7,60	16,84		59,08	
78	7,70	17,49		57,36	
79	7,80	18,14		55,58	
80	7,90	18,79		53,74	
81	8,00	19,43		51,83	
82	8,10	20,08		49,85	
83	8,20	20,73		47,81	
84	8,30	21,38		45,70	
85	8,40	22,03		43,53	
86	8,50	22,67		41,30	
87	8,60	23,32		39,00	
88	8,70	23,97		36,63	

Node	x (m)	V _g (kN)	V _d (kN)	M _g (kN.m)	M _d (kN.m)
89	8,80	24,62		34,20	
90	8,90	25,27		31,71	
91	9,00	25,91		29,15	
92	9,10	26,56		26,53	
93	9,20	27,21		23,84	
94	9,30	27,86		21,09	
95	9,40	28,50		18,27	
96	9,50	29,15		15,39	
97	9,60	29,80		12,44	
98	9,70	30,45		9,426	
99	9,80	31,10		6,349	
100	9,90	31,74		3,207	
101	10,00	32,39		0	

IV - Ultimate Limit States at final stage

IV.1 - Summary of the criteria

- Resistance of sections

Resistance to bending moment (1) :	Γ_M	= 0,216	(N45/ULS-01)	✓
Resistance to shear force :	Γ_V	= 0,062	(N1/ULS-01)	✓
Resistance to MV interaction :	Γ_{MV}	= 0,216	(N45/ULS-01)	✓

(1): Class 1 or 2 cross-sections: plastic design

(2): Shear plastic resistance

- Connection

Degree of shear connection ($M > 0$) :	η	= 0,704	✓
Minimum degree of connection :	η_{min}	= 0,55	

IV.2 - ULS Combinations checkings

Combi	Γ_M	Γ_V	Γ_{MV}
ULS-01	0,216 (N45)	0,062 (N1)	0,216 (N45)

IV.3 - Additional parameters

- Shear buckling

Web slenderness :	h_w/t_w	= 39,2
Limit of web slenderness :	$72\varepsilon_w/\eta$	= 58,6
Where :	ε_w	= 0,81
	η	= 1,00

$h_w/t_w > 72\varepsilon_w/\eta$: The resistance to shear buckling is not required [EN 1993-1-5:2006, 5.1 (2)]

IV.4 - Transverse reinforcements

Number of transverse reinforcement layers :	1
Minimum transverse reinforcement :	$\rho_{t,min}$ = 0,0716 %
Potential areas of shear failure :	a-a

- Shear failure area : a-a

Span	Zone	n_r	$\tau_{Ed,a-a}$ (MPa)	$\theta_{f,min,a-a}$ (°)	$\theta_{f,a-a}$ (°)	$\Gamma_{sf,a-a}$	$(A_{sf}/s_f)_{a-a}$ (cm ² /m)
1	1	1	1,86	27,00	27,00	0,63	1,35

- Shear failure area : c-c

Span	Zone	n_r	$\tau_{Ed,c-c}$ (MPa)	$\theta_{f,min,c-c}$ (°)	$\theta_{f,c-c}$ (°)	$\Gamma_{sf,c-c}$	$(A_{sf}/s_f)_{c-c}$ (cm ² /m)
1	1	1	1,32	27,00	27,00	0,44	2,71

V - Ultimate Limit States at construction stage

V.1 - Summary of the criteria

- Resistance of sections

Resistance to bending moment (1) :	Γ_M	= 0,363	(N51/ULS-C-01)	
Resistance to shear force (2) :	Γ_V	= 0,062	(N1/ULS-C-01)	
Resistance to MV interaction :	Γ_{MV}	= 0,363	(N51/ULS-C-01)	

(1): Class 1 or 2 cross-sections: plastic design
(2): Shear plastic resistance

- Resistance of the beam

Resistance to lateral torsional buckling :	Γ_{LT}	= 1,96	(S2/ULS_1)	
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V.2 - ULS Combinations checkings

Combi	Γ_M	Γ_V	Γ_{MV}	Γ_{LTb}
ULS-C-01	0,363 (N51)	0,062 (N1)	0,363 (N51)	1,956 (N2)

Combi	α_{cr}	M_{cr}
ULS-C-01	0,565	45,79

VI - Serviceability Limit States

VI.1 - Deflections

- Under elementary load cases

Assuming full shear interaction

Load case	Deflections
Self-weight (G1)	35,6 mm (L/281)
Shrinkage of the slab (SHC)	10,5 mm (L/951)

Taking in to account the slip of the connection

Load case	Deflections
Self-weight (G1)	35,6 mm (L/281)
Shrinkage of the slab (SHC)	11,4 mm (L/879)

- Under SLS combinations

Assuming full shear interaction

Combination	Deflections
SLS-01	46,1 mm (L/217)

Taking in to account the slip of the connection

Combination	Deflections
SLS-01	47,0 mm (L/213)

VI.2 - Eigen frequencies

Eigen frequency for the combination G1 : $f = 5,86 \text{ Hz}$

VI.3 - Assessment of floor vibrations

Usage of the floor structure	Office buildings
Structural damping	D1 = 1 %
Damping for furniture	D2 = 2 %
Damping for finishing	D3 = 0 %
Total damping	D = 3 %
Combination of masses for eigenfrequency	G + 0,1 Q1
Eigen frequency	5,86 Hz
Modal mass	2446 kg
Effective value of velocity	OS-RMS ₉₀ = 7,4 m/s
Class of perception	E
Comfort assessment	Critical

